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$$\begin{aligned} & \int \sqrt{x^2 + 1} dx \\ & \stackrel{\sqrt{x^2+1}=x+t}{=} \int \left(-\frac{t^2 - 1}{2t} + t \right) \left(-\frac{t^2 + 1}{2t^2} \right) dt \\ & = \int \left(\frac{t^2 + 1}{2t} \right) \left(-\frac{t^2 + 1}{2t^2} \right) dt \\ & = - \int \frac{(t^2 + 1)^2}{4t^3} dt \\ & = -\frac{1}{4} \cdot \left(\int \frac{t^4 + 2t^2 + 1}{t^3} dt \right) \\ & = -\frac{1}{4} \cdot \left(\frac{t^2}{2} + 2 \cdot \ln |t| - \frac{1}{2t^2} \right) + c \\ & = -\frac{(\sqrt{x^2 + 1} - x)^2}{8} - \frac{\ln |\sqrt{x^2 + 1} - x|}{2} + \frac{1}{8(\sqrt{x^2 + 1} - x)^2} + c \end{aligned}$$

References